Remote Sensing Collaborative Classification Using Multimodal Adaptive Modulation Network

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***Abstract*— With the development of remote sensing technology, more and more data sources are available for landcover classification tasks, such as hyperspectral images (HSIs), light detection and ranging (LiDAR) data, and synthetic aperture radar (SAR) data. Due to the unique information carried by different sources, the collaborative use of multiple remote sensing data has become a key research direction in landcover classification tasks. Most existing methods are only capable of dealing with two types of remote-sensing images, limiting the potential for the collaboration of more sources. Also, the traditional feature fusion method uses addition or concatenation manner to integrate information, which makes it difficult to make full use of complement characteristics between modalities. As a remedy, we propose a novel multimodal adaptive modulation network (MAMNet), for landcover classification tasks using multimodal remote sensing data. First, the cross-modal interacting module (CIM) is utilized for information absorption between modalities. The feature representation is enhanced, and the modality-specific information is preserved. Second, the modal attention layer (MAL) is designed for multimodal feature fusion. Softmax attention is utilized to eliminate redundant information among the three modal features. Finally, an adaptive multimodal margin loss (AMM loss) is proposed to balance the consistency and diversity of multimodal features. It encourages adjustable decision margins between sources, which enables the model to better utilize complementary information between modalities and partially avoids model-overfitting by defining a more difficult learning target. Experimental results on two benchmark remote sensing datasets show the effectiveness of the proposed method compared with several state-of-the-art approaches.**

***Index Terms*— Adaptive multimodal margin loss (AMM loss), feature fusion, modal attention, multimodal classification, multisource remote sensing.**

I. Introduction

REMOTE sensing images have the characteristics of a large observation range and fast information acquisition. They can provide important information services for resource exploration, environmental monitoring, land surveying, and other works [1], [2]. With the development of remote sensing technology, more and more types of remote sensing data are available to improve the accuracy of landcover classification task [3], [4], [5]. Among them, hyperspectral images (HSIs) are the most widely used data type for land classification, because they carry abundant spectral information and can recognize subtle spectral differences among materials [6], [7]. The research results on HSI are quite rich, for example, in [8], a cycle-consistency unmixing network (CyCU-Net) was proposed to solve the problem of losing material information in traditional autoencoders for hyperspectral unmixing tasks. Specifically, a self-perception loss consisting of two spectral reconstruction terms and one abundance reconstruction term is proposed to achieve cycle consistency. To solve the generalized bilinear model (GBM) for nonlinear hyperspectral unmixing tasks, Gao et al. [9] proposed an effective method using low-rank abundance maps and nonlinear interaction abundance maps. Su et al. [10] developed a method based on normalized spectral clustering with kernel-based learning (NSCKL) for HSI classification problems, in which a kernel-based iterative filter (KIF) was designed to make connections between pixels. In [11], a novel blind-spot self-supervised learning network (BS3LNet) was proposed for hyperspectral anomaly detection (HAD) problems. The network reconstructs HSI using a blind-spot architecture, that is to say, when reconstructing a pixel, its receptive field ignores the pixel itself and only uses information from adjacent pixels. Later in [12], a chessboard topology-based anomaly detection (CTAD) method was proposed for HAD problems, which can simultaneously mine high-dimensional data features and dissect HSI adaptively. For unsupervised multispectral-aided hyperspectral image super-resolution (MS-aided HS-SR) tasks, Gao et al. [13] proposed an enhanced unmixing-inspired unsupervised network with attention-embedded degradation learning (EU2ADL) to promote the feature representation ability of unsupervised networks. The method uses two coupled autoencoders to decompose the input image into abundances and corresponding endmembers, and uses a mixed model-constrained loss that includes a perceptual abundance term and a degradation-guided term to train the network to eliminate image distortion. However, HSI is prone to the phenomenon of inconsistency between spectrum and objects, which affects classification performance. Meanwhile, light detection and ranging (LiDAR) data have no advantage

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in spectral detection, but they are provided with excellent structural and information acquisition ability [14], [15], which is valuable for better describing the same scene obtained only by optical sensors. Other available data types, including multispectral images (MSIs) and synthetic aperture radar (SAR) data, possess different advantages and disadvantages in the landcover classification task. Therefore, reasonable utilization of monitoring information from various remote sensors has a positive impact on improving classification performance.

In the field of multisource remote sensing image classification, the commonly used fusion strategies are mainly divided into three categories, namely data fusion, feature fusion, and decision fusion. Among them, feature fusion is the most commonly used fusion manner. More specifically, in [16], a generalized graph-based fusion method was utilized to fuse and reduce the dimensionality of morphological profiles (MPs) features acquired from HSI and LiDAR data. Khodadadzadeh et al. [17] pointed out that the simple stacking and addition of multisource features may lead to information redundancy, and they proposed a feature fusion method based on SVM, achieving state-of-the-art classification performance. In [18], extinction profiles (EPs) were utilized to fuse information of LiDAR data and HSI. Later in [19], a fusion method based on the discriminant graph was utilized to minimize interclass similarity and maximize interclass distance, thus extracting heterogeneous features of HSI and LiDAR. In [20], fully connected (FC) deep neural networks (DNNs) were used to fuse heterogeneous features of HSI, MSI, and LiDAR data, and logistic regression was used to obtain the final classification results. Xu et al. [21] further designed a dual-tunnel convolutional neural network (CNN) to extract spectral and spatial features of HSI, and a CNN with cascade blocks was developed to extract multiscale features of LiDAR data. More recently, Hong et al. [22] learned joint graph structures from multiple remote sensing data and proposed an optimization strategy based on the ADMM method to implement the solving process. Zhang et al. [23] applied patch-to-patch mapping of HSI and LiDAR data to deeply fuse their multiscale features. In [24], isotropic filter banks or convolutional kernels were used to extract spatial features of remote sensing images, and Fourier transform was used to extract frequency domain features, finally obtaining spatial frequency-invariant features for classification. Hong et al. [25] proposed a novel cross-modal transfer learning framework for large-scale MSI and SAR data classification. In order to further enhance the discriminative ability of features, the self-attention mechanism and weights-sharing settings are widely introduced into feature fusion methods. In [26], the self-attention mechanism was utilized to learn spectral features of HSI, and the cross-attention mechanism was utilized to interact with LiDAR features and spatial features of HSI. Hong et al. [27] proposed a novel S2FL model to decompose remote sensing data into a modal sharing part and a modal heterogeneity part, to fuse multimodal features more effectively. In addition, some models based on attention [28], adversarial learning [29], and transformer [30] have been proposed for feature fusion. Generally speaking, these feature fusion methods have been proven to be effective in improving multimodal classification accuracy, yet they can only handle two-modal-based input collaboration. Also, their ability to handle heterogeneous characteristics and parsing complementary information remains limited.

On the other side, data fusion and decision fusion-based methods have also achieved good classification performance. Data fusion transforms the multisource classification problem into a single-source classification problem by directly fusing multisource remote sensing images. For example, Byun [31] integrated optical and SAR images through wavelet domain integration. Later in [32], the Gram–Schmidt algorithm was utilized to fuse high-resolution SAR data and optical images, and SVM was used for subsequent classification. Dian et al. [33] used a well-trained CNN for gray image denoising to regularize the HSI-MSI fusion problem. Decision fusion typically performs feature extraction and classification on a portion of data sources separately and then votes on the classification results of multiple classifiers to obtain the final classification result. For example, Liao et al. [34] fused HSI features and LiDAR features using graph-based methods, and then used SVM and voting strategy to obtain the final classification result. In [35], an active learning (AL) strategy based on multi-kernel learning was adopted to optimize the kernels of multiple data sources, and decision fusion was implemented to make the final decision based on the probabilistic outputs. Later Zhong et al. [36] utilized weighted voting to obtain the classification result, and the weights of each source's classification map are optimized by the differential evolution algorithm based on the self-adaptive strategy. More recently, Hang et al. [37] proposed two coupled CNNs to extract HSI and LiDAR features, and multiple feature fusion methods were attempted to enhance the features. For generating the final classification results, a weighted summation strategy was adopted based on the accuracy of the outputs. Although data fusion and decision fusion methods have been able to obtain promising results, their ability in characterizing heterogeneous information remains inadequate, making it difficult for the model to obtain discriminative classification criteria. Therefore, as compared to the case of feature fusion methods, data fusion and decision fusion are relatively less investigated by RS researchers.

In conclusion, the aforementioned classification methods are difficult to extend the classification scenes with more than two remote sensing sources. If we can enable the network to extract features from more remote sensing data sources and overcome problems such as overfitting caused by excessive information, we can enable the network to learn more information about the target area, which is more conducive to achieving better classification accuracy than methods based on two-source remote sensing data. Besides, the traditional feature fusion methods often focus on the consistency of the multimodal feature, ignoring the potential complementarity. Specifically, for multisource images, some modalities share the common information, while some have their own specific information. In general, consistency and complementarity are mutually exclusive and difficult to balance. To achieve effective feature extraction, a proper tradeoff between the two strategies is essential, which encourages the network to emphasize feature discrimination.

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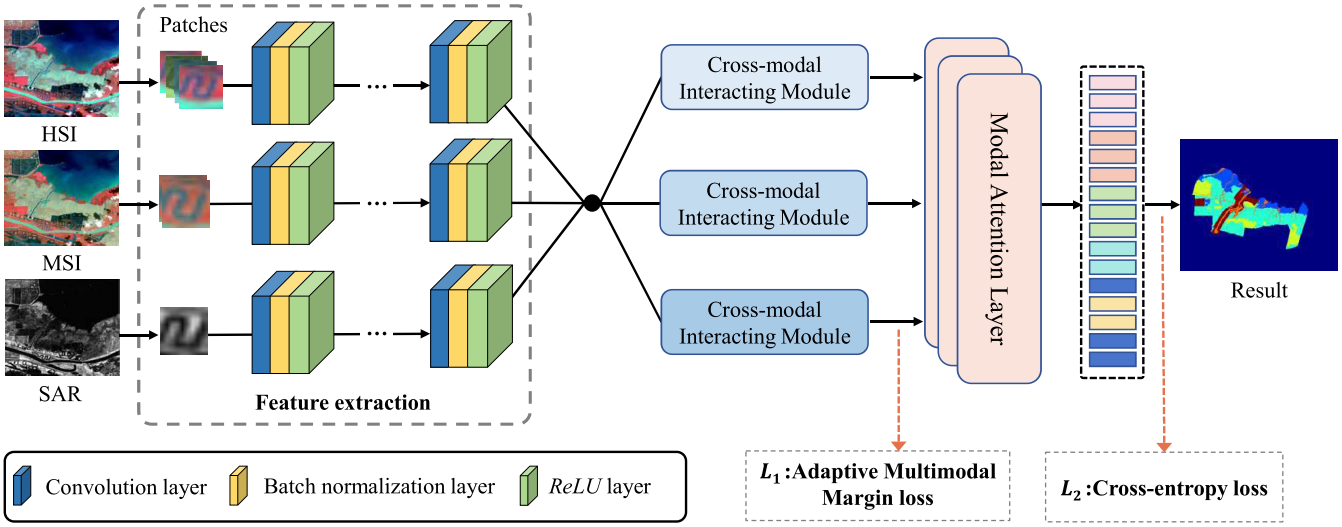


Fig. 1. Flowchart of the proposed MAMNet. First, the feature extraction network is utilized for feature embedding, and then a set of CIM modules and MALs are cascaded for feature interaction and redundancy elimination. Finally, the combination of AMM loss and cross-entropy loss is utilized for optimization while maintaining diversity and complementarity of multimodal features.

To address the above problems, we propose a novel multimodal adaptive modulation network (MAMNet) that balances the diversity and consistency of features while extracting features from multiple kinds of remote sensing data, thereby maximizing the positive impact of the complementary information on model performance. Specifically, after independently extracting features from multimodal data sources, the cross-modal interacting module (CIM) [38] is utilized to fully absorb the complement merits from other modalities, in which the channel attention (CA) module is utilized to eliminate the redundant information. Moreover, to enhance meaningful modality information while fusing features, a novel modal attention layer (MAL) is designed. Multimodal feature vectors are re-encoded using the tanh function, and Softmax attention is utilized between modalities. In this way, important modality information is preserved, and redundant information is eliminated. In addition, a novel loss function named adaptive multimodal margin loss (AMM loss) is proposed. To enhance the learning efficiency for modal complementary information, the diversity and consistency of multimodal features are adaptively balanced based on the input modalities during the training process. The best $L2$ distances between multimodal features are automatically learned by the proposed network so that the network can adapt to any combinations of input modalities. Experiments on two datasets show that the proposed MAMNet has a competitive classification accuracy compared to several state-of-the-art classification methods.

The main contributions of the proposed method are summarized as follows.

1) A CIM module is designed to extract complementary advantages from multi-source remote sensing data. Specifically, the CIM of each modal branch adaptively absorbs dominant information from other modalities, while redundant information is suppressed through attention modules.
2) After the CIM module, a novel AMM loss is designed, which balances the diversity and consistency of multimodal features by adjusting the $L2$ distance among them. Diversity means enhanced complementarity of multimodal features, while consistency is aimed at avoiding semantic conflicts caused by feature heterogeneity.
3) A MAL is developed for feature integration. Different from the CA module, the MAL generates one attention weight for each branch, avoiding the destruction of the diversity and consistency of multimodal features.

The rest of this article is organized as follows. In Section II, the proposed MAMNet and optimization procession is described in detail. Section III provides the description of two three-modal datasets and experimental results. Finally, Section IV concludes this article.

II. Proposed Classification Framework

Take three-modal data input as an example, the proposed MAMNet is shown in Fig. 1. First, four layers of Conv $\rightarrow$ BN $\rightarrow$ ReLU network are used to extract features of the three modalities separately and unify the number of channels of the three features as c. Second, the CIM [38] is used to fully absorb information from the other two modalities, thereby enhancing the network's ability to utilize complementary information. The fused multimodal features are fed into the AMM loss function to make the distances of the features more suitable for classification. Third, the MAL is proposed to learn the optimal linear combination of three modal features, and fuse the three modal features into one final feature at the same time. Finally, the final feature is input into the FC layer to obtain the final classification results.

A. Feature Extraction Network

Multimodal remote sensing data has different advantages in terms of information dimensions due to differences in imaging mechanisms. Generally speaking, the advantages of spectrum in HSI, the spatial texture of MSI, the elevation of LiDAR, and the all-weather observation characteristics of SAR provide

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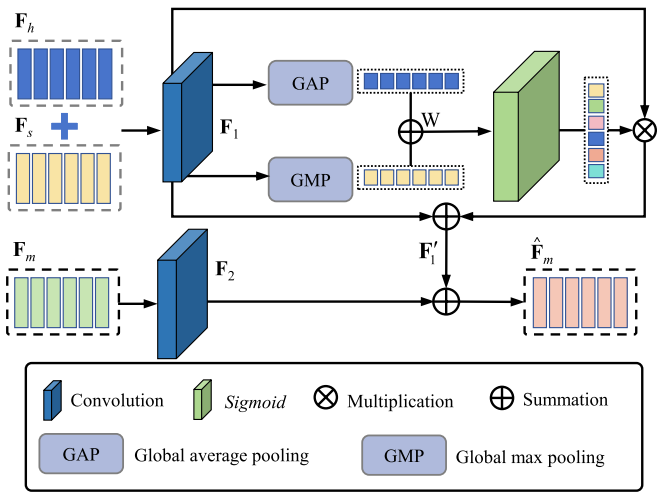


Fig. 2. Flowchart of the CIM.

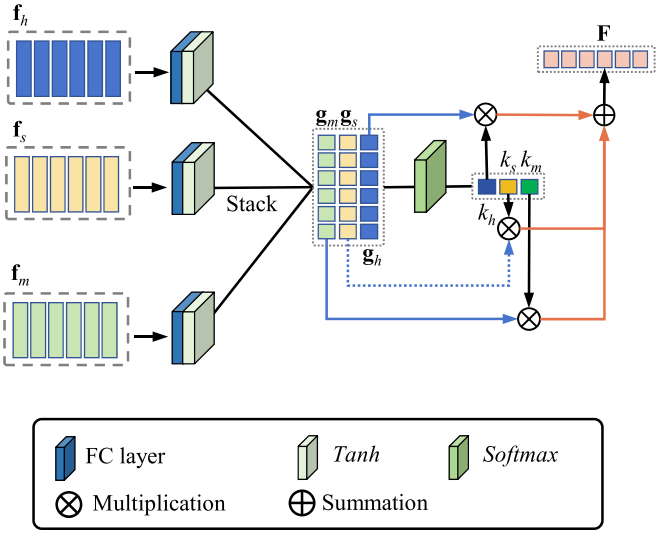


Fig. 3. Flowchart of the proposed MAL.

complementary information for collaborative interpretation. Therefore, as shown in Fig. 1, a three-stream Conv → BN → ReLU network is utilized to extract three-modal features separately. Parameters are not shared between the three branches so the modality-specific information is best reserved. Besides, to facilitate the interaction of the three features in subsequent steps, the output channels of each Conv → BN → ReLU layer are all set to c.

B. Cross-Modal Interacting Module

Existing feature fusion-based methods for multisource remote sensing image classification tasks mainly focus on how to integrate cross-modal information. However, such a strategy is prone to ignore modality-specific information, so complementarity is often compromised [39]. To address this problem, the CIM is designed to integrate the feature from other modalities through summation operation, achieving both modality-specific information preservation and information absorption from other modalities. In addition, the CA module is employed to eliminate redundant information of the fused feature.

The detailed structure of the CIM is shown in Fig. 2. The input features of the CIM, denoted as $\mathbf{F}_h, \mathbf{F}_s, \mathbf{F}_m \in \mathbb{R}^{h\times w\times c}$, are generated by the above-mentioned feature extracting operations from the input image patches

$$\begin{aligned}\mathbf{F}_h &= \mathrm{CBL}_4(\mathbf{p}_h)\\ \mathbf{F}_s &= \mathrm{CBL}_4(\mathbf{p}_s)\\ \mathbf{F}_m &= \mathrm{CBL}_4(\mathbf{p}_m)\end{aligned} \tag{1}$$

where $\mathrm{CBL}_4(\cdot)$ stands for the processing of four layers of Conv → BN → ReLU network, and $\mathbf{p}_h, \mathbf{p}_s, \mathbf{p}_m$ denote the image patches. If we want to let $\mathbf{F}_m$ absorb information from $\mathbf{F}_h$ and $\mathbf{F}_s$, we first add $\mathbf{F}_h$ and $\mathbf{F}_s$ pixel by pixel, and then feed them to a convolution layer to obtain the fused feature $\mathbf{F}_1$. Meanwhile, $\mathbf{F}_m$ is fed to a convolution layer to obtain the extracted feature $\mathbf{F}_2$.

The multimodal features obtained from the feature extraction network have a limited receptive field, and due to the characteristics of convolution operation, the channel correlation and spatial correlation in the output feature are entangled [40], which is difficult to use by the network. In order to enhance the meaningful bands of $\mathbf{F}_1$, the CA module is utilized to generate the attention weights. Specifically, global max pooling (GMP) and global average pooling (GAP) are first conducted on $\mathbf{F}_1$ to obtain the channel weight coefficient matrix W. Then, the enhanced features $\mathbf{F}_1'$ are obtained as follows:

$$\mathbf{F}_1' = \mathbf{F}_1 \times \sigma(W) + \mathbf{F}_1 \tag{2}$$

where $\sigma(\cdot)$ is the Sigmoid function, W is the channel weight coefficient matrix, and $\times$ denotes the channel-wise multiplication. Finally, the fused feature map $\hat{\mathbf{F}}_m$ is obtained through pixel-wise summation

$$\hat{\mathbf{F}}_m = \mathbf{F}_1' + \mathbf{F}_2. \tag{3}$$

Similarly, we can operate on the other two features $\mathbf{F}_s$ and $\mathbf{F}_h$ to obtain the fused features $\hat{\mathbf{F}}_s$ and $\hat{\mathbf{F}}_h$ so that all features from the three modalities can absorb useful information to provide more promising classification results.

C. Modal Attention Layer

After feature interaction through CIM, the multimodal features need to be fused before input to the FC layer for final classification. A novel MAL is designed to minimize information redundancy.

The proposed MAL is shown in Fig. 3. First, the feature maps $\hat{\mathbf{F}}_h, \hat{\mathbf{F}}_s, \hat{\mathbf{F}}_m \in \mathbb{R}^{h\times w\times c}$ will be reshaped to $\mathbf{f}_h, \mathbf{f}_s, \mathbf{f}_m \in \mathbb{R}^{n\times 1}$, $n = c \times h \times w$, where c, h, and w denote the number of channels, height, and width of the three feature maps, respectively. Usually, the concatenation or addition operation is used to fuse the multimodal features. However, because all modalities express the same scene, the concatenation operation is too simple to make full use of the inherent correlation between the modalities [41]. Also, using the addition operation is still not an optimal choice, because multimodal feature maps

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do not share statistical properties [42]. Thus, this strategy is not promising for effectively utilizing the complementarity of multisource data. Because different data sources have different natures, their contributions to the predictions should be different. To better cope with multiple different data combinations, instead of designing the fusing process manually, we let the model learn how to fuse them optimally.

First, a new feature representation is obtained through an FC layer and the tanh activation function

$$\mathbf{g}_h^{1\times m} = \tanh\left(\mathbf{f}_h^{1\times n} W_h^{n\times m}\right) \tag{4}$$

where W_h denotes the weight matrix of the FC layer, and $\mathbf{g}_h$ denotes the new feature representation. The same operation will be applied to $\mathbf{f}_s$ and $\mathbf{f}_m$, thus obtaining new feature vectors $\mathbf{g}_s$ and $\mathbf{g}_m$. We represent the fused feature as a linear combination of multimodal features, and the contribution of each modal to the fused feature as a coefficient. Therefore, the optimal strategy of fusing is obtained by computing the coefficients. To do this, $\mathbf{g}_h$, $\mathbf{g}_s$, and $\mathbf{g}_m$ are stacked, and the coefficients are computed through Softmax function

$$[k_h, k_s, k_m] = \text{Softmax}\left([\mathbf{g}_h, \mathbf{g}_s, \mathbf{g}_m]\right) \tag{5}$$

where k_h, k_s, k_m are the contribution coefficients. At last, the final feature is represented as a linear combination of three modal features

$$\mathbf{F} = k_h \cdot \mathbf{g}_h + k_s \cdot \mathbf{g}_s + k_m \cdot \mathbf{g}_m \tag{6}$$

where $\cdot$ denotes the pixel-wise multiplication. In this way, important information in the modality will be preserved and efficiently represented in the final features.

D. Adaptive Multimodal Margin Loss

To make full use of the complement information of these modalities remote sensing data, a proper tradeoff needs to be made between the diversity and consistency of the input features. Here, diversity means enhanced complementarity of multimodal features, while consistency is aimed at avoiding semantic conflicts caused by feature heterogeneity. As far as we know, different modality carries different types and amounts of information. Therefore, when the information contained in the two modalities is similar, the extracted features should tend to be similar and vice versa. To make the proposed model adaptable to various combinations of input modalities, we propose a novel loss function named AMM loss, which enables the network to adaptively learn how to balance the diversity and consistency of input modalities.

The overall scheme of the proposed AMM loss is shown in Fig. 4. First, the inspiration for the proposed AMM loss came from the standard triplet margin loss [43], which is formulated as

$$T\left(\hat{\mathbf{F}}_h, \hat{\mathbf{F}}_s, \hat{\mathbf{F}}_m\right) = \max\{d\left(\hat{\mathbf{F}}_h, \hat{\mathbf{F}}_s\right) - d\left(\hat{\mathbf{F}}_h, \hat{\mathbf{F}}_m\right) + \text{margin}, 0\} \tag{7}$$

where $\hat{\mathbf{F}}_h$, $\hat{\mathbf{F}}_s$, and $\hat{\mathbf{F}}_m$ represent the multimodal features generated by CIM, $d(\cdot,\cdot)$ denotes the Euclidean distance between two features, and margin is a hyperparameter. The appropriate distances between features can balance their diversity and consistency, thereby achieving complementarity. The triplet margin loss considers $\hat{\mathbf{F}}_h$ as an anchor, during the training process, it shortens the distance between anchor and positive ($\hat{\mathbf{F}}_s$), and widen the distance between anchor and negative ($\hat{\mathbf{F}}_m$), so that the consistency between the anchor point and the positive sample will be strong, while the diversity between the anchor point and the negative sample will be strong. However, we want to make our model have the same good classification performance for three modal inputs in different orders, but the triplet margin loss function cannot produce the same output result when exchanging input orders, and more importantly, the feature distribution that can achieve optimal classification performance may not satisfy the distribution learned through the triplet margin loss, so it cannot be a suitable loss function for our proposed method.

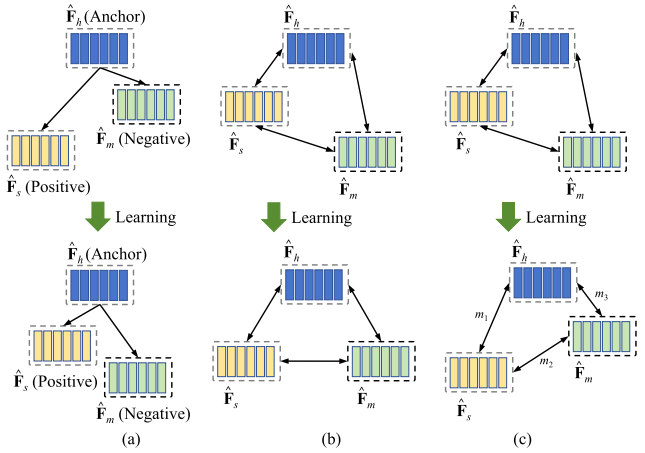


Fig. 4. Schematic of (a) triplet margin loss, (b) improved triplet loss, and (c) proposed AMM loss.

Following this, the idea came to the improved triplet loss to make the distribution of the three-modal features approaches an equilateral triangle as follows:

$$\begin{aligned} L_1'\left(\hat{\mathbf{F}}_h, \hat{\mathbf{F}}_s, \hat{\mathbf{F}}_m\right) = {} & T\left(\hat{\mathbf{F}}_h, \hat{\mathbf{F}}_s, \hat{\mathbf{F}}_m\right) + T\left(\hat{\mathbf{F}}_h, \hat{\mathbf{F}}_m, \hat{\mathbf{F}}_s\right) \\ & + T\left(\hat{\mathbf{F}}_s, \hat{\mathbf{F}}_h, \hat{\mathbf{F}}_m\right) + T\left(\hat{\mathbf{F}}_s, \hat{\mathbf{F}}_m, \hat{\mathbf{F}}_h\right) \end{aligned} \tag{8}$$

where the first two terms is to let $d(\hat{\mathbf{F}}_h, \hat{\mathbf{F}}_s)$ approaches $d(\hat{\mathbf{F}}_h, \hat{\mathbf{F}}_m)$, and the last two terms is to let $d(\hat{\mathbf{F}}_h, \hat{\mathbf{F}}_s)$ approaches $d(\hat{\mathbf{F}}_s, \hat{\mathbf{F}}_m)$. Therefore, a tradeoff between diversity and consistency of the multimodal features is realized, thus achieving complementarity. However, this loss function cannot control the value of the distance between the multimodal features. That is to say, the distance between the features may be too large, so that the consistency between features may be compromised.

To address the above problem, the AMM loss is developed to directly control the distance between multimodal features, which is formulated as follows:

$$\begin{aligned} & L_1\left(\hat{\mathbf{F}}_h, \hat{\mathbf{F}}_s, \hat{\mathbf{F}}_m\right) \\ & = \max\left(|m_1 - d\left(\hat{\mathbf{F}}_h, \hat{\mathbf{F}}_s\right)|, |m_2 - d\left(\hat{\mathbf{F}}_h, \hat{\mathbf{F}}_m\right)|, |m_3 - d\left(\hat{\mathbf{F}}_s, \hat{\mathbf{F}}_m\right)|\right) \end{aligned} \tag{9}$$

where $|\cdot|$ denotes the absolute value calculation, m_1, m_2, m_3 are learnable parameters, and $d(\cdot,\cdot)$ denotes

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES IN TWO DATASETS

Augsburg dataset				Yellow River Estuary dataset			
Class		Number of Sample		Class		Number of Sample	
No.	Name	Training	Testing	No.	Name	Training	Testing
1	Forest	146	13361	1	Spartina Alterniflora	735	39784
2	Residential Area	264	30065	2	Suaeda Salsa	2519	118213
3	Industrial Area	21	3830	3	Tamarix Forest	1297	35216
4	Low Plants	248	26609	4	Tidal Creek	529	15673
5	Allotment	52	523	5	Mudflat	702	24592
6	Commercial Area	7	1638				
7	Water	23	1507				
Total		761	77533	Total		5782	233478

the $L2$ distance between two features. Before the training process, m_1, m_2, m_3 are all set to 1, that is to say, we first assume that the differences between each modality are the same, and the optimal $L2$ distances between them are all 1. Therefore, if the distance between any two modalities is not equal to 1, the value of the loss function can be optimized during training, and the distance with the maximum absolute difference from 1 is always the first to be optimized. However, as the training progresses, m_1, m_2, and m_3 will also be optimized based on the value of the classification loss, allowing the network to adaptively adjust the optimal distance between modalities, ultimately converging to the distance that is most beneficial for classification. In this way, the optimal distance and actual distance between modalities will be optimized simultaneously, achieving a balance between consistency and diversity among modalities, so that complementarity can be best achieved.

After feature extraction and fusion which are introduced above, a FC layer is utilized to obtain the final classification labels. The cross-entropy loss is utilized to measure the difference between the output labels and the ground truth (GT) labels. The cross-entropy loss is calculated as follows:

$$L_2 = \frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} -y_i^c \log \ p_i^c \tag{10}$$

where N denotes the batch size, and y_i^c means that if the label of the ith data is equal to c, than $y_i^c = 1$, otherwise $y_i^c = 0$. Finally, the total loss function of MAMNet is denoted as follows:

$$L = \lambda L_1 + L_2 \tag{11}$$

where λ is a hyperparameter, and λ is set to 2.

III. EXPERIMENTAL RESULTS AND DISCUSSIONS

In this section, the model configuration and performance are analyzed through experiments. The considered datasets are first described, and then we determine the optimal hyperparameters through experiments. Comparative experiments with relevant methods are conducted to verify the performance of the proposed method.

A. Description of Dataset

The first dataset is the Augsburg dataset, including HSI, SAR, and digital surface model (DSM) data, which are captured by the HySpex sensor, C-band SAR sensor of Sentinel-1 satellite, and DLR-3 K system, respectively. It is located in an urban area of Augsburg, Germany. The land features in this area are relatively scattered and have both artificial and natural categories. The size of the images of all three modalities are all 332 × 485 pixels. HSI images have 180 channels with a wavelength range from 400 to 2500 nm, and SAR images and DSM images have four and one channels, respectively.

The second dataset is the Yellow River Estuary dataset which contains HSI, MSI, and SAR images. The HSI and MSI data were taken by the ZiYuan1-02D satellite, and the SAR data were captured by the Sentinel-1 satellite. It is located in a wetland scene in Shandong, China. The land features in this area are relatively continuous, containing only natural categories and not artificial categories, making it relatively difficult to distinguish between classes and to depict the boundaries between each category. The numbers of channels are 166, 8, and 3, respectively, and the sizes of the images are all resized to 960 × 1170.

Table I lists the number of training and testing samples from the region of interest. The illustration of the two datasets is shown in Fig. 5. For the Augsburg dataset, the GT map is obtained by using the OpenStreetMap data. For the Yellow River Estuary dataset, the GT map is generated by annotation during the field investigation. Data augmentation strategies are used to enrich the dataset and weaken overfitting. In our experiments, data augmentation with noise and 90-degree rotation were used on both datasets. During training, 80% of the training samples are for training, and the rest are for validation. The MAMNet utilizes stochastic gradient descent (SGD) as the optimizer, with a learning rate of 0.0005, a momentum of 0.9, and a weight decay of 0.0005. The number of training epochs is set to 300, and batch size is set to 100. The whole MAMNet is implemented on Pytorch, and the experiment environment is an Nvidia GeForce GTX 1660 GPU.

B. Parameter Tuning of the Proposed MAMNet

In this section, different hyperparameter configurations are measured by classification metrics. The ablation experiments are conducted to prove the effectiveness of the core modules

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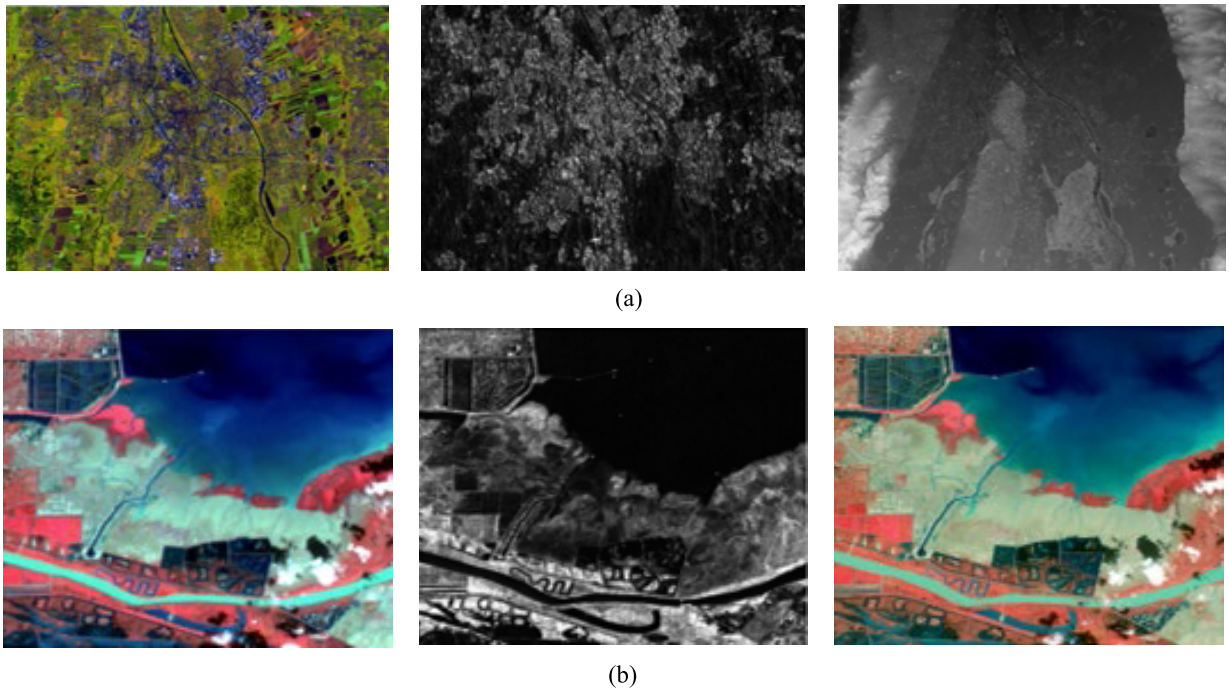


Fig. 5. Three-modal remote sensing datasets. (a) Augsburg dataset. (b) Yellow River Estuary dataset.

TABLE II
RELATIONSHIP BETWEEN IMAGE PATCH SIZE AND CLASSIFICATION PERFORMANCE IN AUGSBURG DATASET

Class	Patch size				
	5	7	9	11	13
1	**95.94**	94.30	93.88	94.12	93.74
2	95.71	96.91	97.42	97.83	**98.48**
3	51.62	65.01	**71.62**	58.25	53.97
4	88.56	**92.63**	88.14	86.48	83.72
5	**60.23**	54.68	59.85	56.21	56.21
6	**14.04**	10.01	0.00	13.92	9.83
7	45.72	**50.03**	30.52	41.27	45.19
OA(%)	88.18	**90.38**	88.74	88.19	87.21
AA(%)	64.55	**66.23**	63.06	64.01	63.02
Kappa	0.8291	**0.8608**	0.8376	0.8300	0.8168

TABLE III
RELATIONSHIP BETWEEN IMAGE PATCH SIZE AND CLASSIFICATION PERFORMANCE IN YELLOW RIVER ESTUARY DATASET

Class	Patch size				
	9	11	13	15	17
1	95.10	94.86	**95.47**	95.00	95.14
2	85.76	89.01	86.18	**90.11**	87.84
3	74.80	81.98	**85.34**	83.33	84.22
4	83.52	81.86	**84.21**	83.74	82.13
5	**81.69**	73.52	75.45	69.88	77.23
OA(%)	85.12	86.83	86.37	**87.36**	87.04
AA(%)	84.17	84.25	**85.33**	84.41	85.31
Kappa	0.7838	0.8062	0.8025	**0.8133**	0.8109

of the proposed framework. Overall accuracy (OA), average accuracy (AA), and Kappa are utilized to measure the classification performance.

1) Analysis of Patch Size: The image patch size is an important parameter in the pixel-wise classification task. If the patch size is too small, the spatial information in a single patch is insufficient. If the patch size is too large, the experimental results exhibit a serious artificial trace, and the computational burden will increase. Besides, for different datasets, the optimal patch size of the same classification model is different. Usually, urban scenes tend to choose smaller image patch sizes, while wetland scenes choose larger ones.

The relationship between image patch size r and classification performance on two datasets is listed in Tables II and III. From Table II, $r = 7$ is optimal for the Augsburg dataset. However, for the Yellow River Estuary dataset, when $r = 15$ achieves the best OA and Kappa coefficient, but $r = 13$ achieves the best AA, and it is optimal for most of the five classes. The main reason for this result is that a larger patch size makes it easy to ignore the features of fragmented land cover. In addition, $r = 15$ makes it hard to extract the fine-grained features, such as tidal creeks. Therefore, we recommend a patch size of 13 for wetland scenes like the Yellow River Estuary, and a patch size of 7 for urban scenes like Augsburg. For most typical scenarios, a patch size of 11 is a good unified solution.

2) Analysis of Hyperparameter λ: In the total loss function of the proposed network, the hyperparameter λ is used to adjust the numerical values of the AMM loss and the cross-entropy loss to achieve the optimal training effect. The

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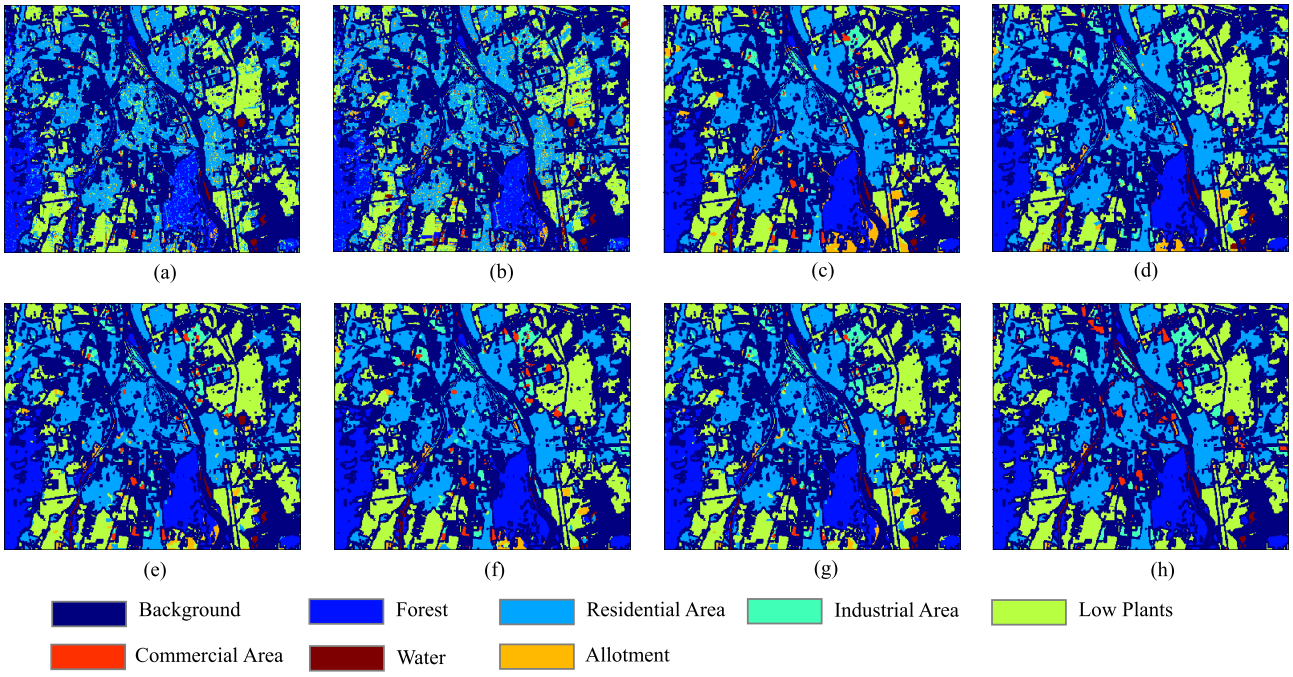


Fig. 6. Classification maps using different methods on the Augsburg dataset. (a) SVM. (b) LBP-ELM. (c) Three-branch CNN. (d) ResNet. (e) CExNet. (f) IEAE. (g) MAMNet. (h) GT.

TABLE IV
RELATIONSHIP BETWEEN λ AND OA IN AUGSBURG DATASET

λ	Overall Accuracy (%)
1	88.82
2	**90.38**
3	88.87
4	87.42

TABLE V
ABLATION STUDIES FOR THE PROPOSED MAMNET ON TWO DATASETS

Index	MAL	CIM	AMM loss	Overall Accuracy (%)	
				Augsburg	Yellow River Estuary
1	w/o	w/o	w/o	86.76	86.02
2	✓	w/o	w/o	87.70	86.80
3	✓	✓	w/o	88.83	86.94
4	✓	✓	✓	**90.38**	**87.36**

relationship between the value of λ and OA on the Augsburg dataset is shown in Table IV. We can see that when λ is equal to 2, the OA is the highest, so it is fixed in further experiments.

3) Ablation Studies: In the proposed MAMNet, three core parts are utilized to extract the discrimination feature from multimodal data, including CIMs, MAL, and AMM loss. Specifically, three CIMs are used to preserve modal-specific information while performing feature interaction. Furthermore, the MAL is proposed to eliminate redundant information while fusing multimodal features. Finally, in order to promote complementarity, the AMM loss is proposed to adaptively adjust the diversity and consistency of multimodal features. To demonstrate the effectiveness of the three core modules, ablation experiments are conducted on two datasets, and the results are listed in Table V. It is observed from indices 1 to 4, that the OA has gradually improved significantly on both datasets as more modules are applied. Therefore, all three core modules have played a significant positive role in the classification performance of the model.

C. Classification Performance

In this section, our proposed MAMNet is compared with some advanced classification models, including SVM [44], LBP-ELM [45], three-branch CNN [21], five-layer ResNet, CExNet [46], and IEAE [38]. In [44], binary SVMs are applied to multiclass remote sensing image classification tasks. The LBP-ELM uses local binary patterns to extract spatial features of remote sensing images and the extreme learning machine as a classifier. The three-branch CNN uses dual-tunnel CNN to extract HSI features, and the cascaded blocks are employed to extract the features from other modalities. The five-layer ResNet is extended to three branches with the same structure. The CExNet employs the channel exchanging strategy after feature extraction to enhance meaningful channels. The IEAE method first makes the interaction between multisource features during the feature extraction stage and then embeds global features into fine-grained local information. The values of classification metrics of all the competitors are shown in Tables VI and VII, where Table VI indicates the results on the Augsburg dataset, and Table VII indicates the results on the Yellow River Estuary dataset.

1) Results on the Augsburg Dataset: The classification maps of the Augsburg dataset are shown in Fig. 6. We can see that the classification maps generated by SVM and LBP-ELM suffer from a lot of noise. This is because these machine

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TABLE VI
CLASSIFICATION METRICS ON THE AUGSBURG DATASET USING DIFFERENT METHODS

No.	SVM [44]	LBP-ELM [45]	Three-branch CNN [21]	ResNet	CExNet [46]	IEAE [38]	MAMNet
1	84.35	86.95	93.27	94.60	96.21	**97.93**	94.30
2	88.14	73.37	97.52	96.24	**97.93**	96.90	96.91
3	20.99	37.99	64.49	**88.67**	52.22	64.49	65.01
4	79.09	91.18	87.12	83.26	88.20	91.30	**92.63**
5	38.43	7.89	**66.16**	60.61	52.39	58.70	54.68
6	8.49	**25.04**	14.22	4.40	0.00	18.01	10.01
7	45.59	**51.91**	48.44	46.52	44.86	20.17	50.03
OA (%)	78.22	74.94	88.66	87.98	88.63	90.14	**90.38**
AA (%)	52.15	53.47	67.32	**67.76**	61.69	63.93	66.23
Kappa	0.6818	0.6442	0.8389	0.8301	0.8346	0.8591	**0.8608**

TABLE VII
CLASSIFICATION METRICS ON THE YELLOW RIVER ESTUARY DATASET USING DIFFERENT METHODS

No.	SVM [44]	LBP-ELM [45]	Three-branch CNN [21]	ResNet	CExNet [46]	IEAE [38]	MAMNet
1	94.30	92.65	94.98	95.62	91.46	**96.21**	95.00
2	83.56	82.79	85.37	85.70	86.46	83.74	**90.11**
3	71.23	77.75	83.54	81.80	84.00	**90.52**	83.33
4	76.56	84.86	83.53	82.27	82.17	**87.46**	84.74
5	71.65	62.02	**79.31**	76.30	77.52	75.05	69.88
OA (%)	81.81	81.77	85.97	85.58	85.71	86.22	**87.36**
AA (%)	79.46	80.01	85.34	84.34	84.32	**86.60**	84.41
Kappa	0.7350	0.7238	0.7978	0.7906	0.7936	0.8030	**0.8133**

learning methods are not good at utilizing spatial information. The three-branch CNN and ResNet largely mistaken low plants as allotment on the Augsburg dataset. Small patches of water are not correctly recognized by ResNet and the IEAE method. The classification maps generated by CExNet and the IEAE method look better, but they are still a little noisy. In contrast, the classification map by MAMNet is closer to the GT map.

It is summarized from Table VI that the proposed MAMNet outperforms other methods in the Augsburg dataset in terms of OA and Kappa coefficient. Specifically, the OA improvements compared with other methods are +12.16%, +15.44%, +1.72%, +2.40%, +1.75%, and +0.24%. It can be observed that machine-learning methods failed to classify classes with large data volumes such as residential area well, and both have low OA. In contrast, deep learning methods have at least 10% improvement in OA. However, the accuracy in the commercial area class by the LBP-ELM is very high. The accuracies of the IEAE method and the proposed method are relatively balanced in classes with more and less training data. Three-branch CNN also has a very good performance on classes that have less training data, and the CExNet performs better on bigger volume classes. Overall, the proposed method achieves the highest OA due to the utilization of multiple effective feature interactions and fusion strategies.

2) Results on the Yellow River Estuary Dataset: The classification maps of the Yellow River Estuary dataset are shown in Fig. 7. The classification maps generated by all methods except for the proposed MAMNet are quite noisy on the left side. Specifically, most of the noise is generated due to misclassification of mudflats. In contrast, the proposed MAMNet generates a classification map that is closest to the GT map.

It is observed from Table VII that the proposed MAMNet achieves the best result on OA and Kappa coefficient in the Yellow River Estuary dataset. The OA improvements of the proposed method with other methods are +5.55%, +5.59%, +1.39%, +1.78%, +1.65%, and +1.14%, respectively. It can be seen from Table I that the training samples of all classes in the Yellow River Estuary dataset are more than those in the Augsburg dataset. In this case, the classification accuracy of each method in each class is relatively balanced. The classification performance of machine-learning methods in the Yellow River Estuary dataset is significantly better than that in the Augsburg dataset. The IEAE method performs the best in most classes, but it fails to effectively classify the larger patches of Suaeda salsa and mudflat. In contrast, the proposed MAMNet effectively classifies Suaeda salsa, with much higher accuracy than other comparison methods.

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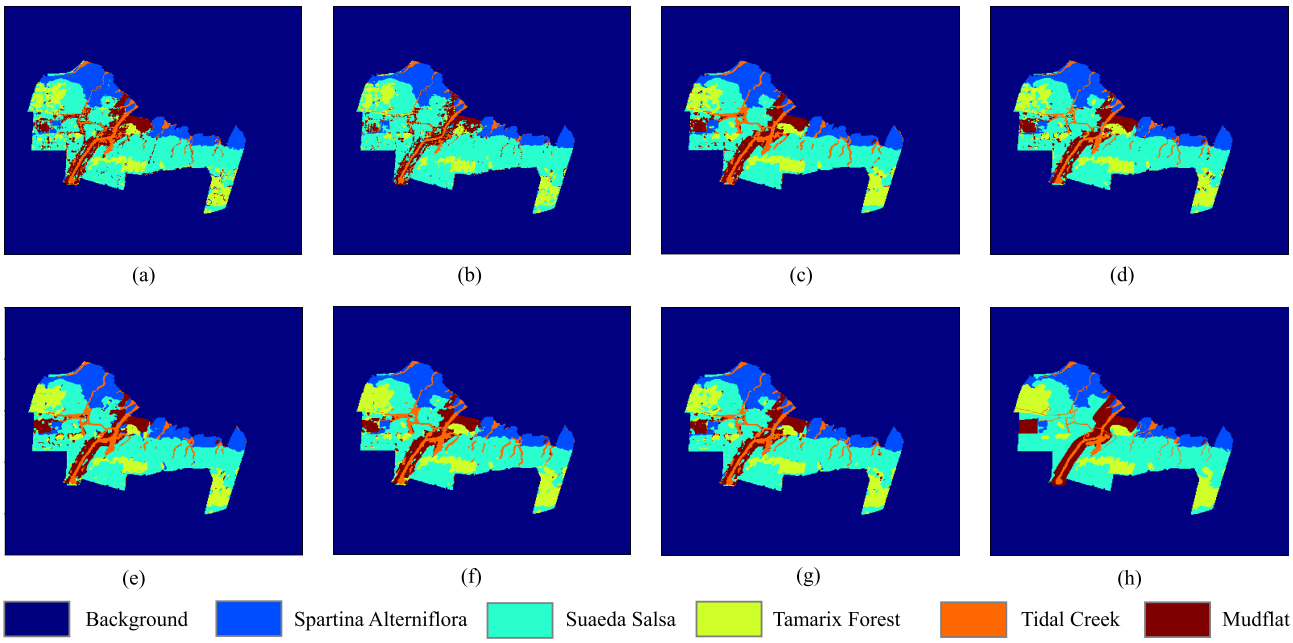


Fig. 7. Classification maps using different methods on the Yellow River Estuary dataset. (a) SVM. (b) LBP-ELM. (c) Three-branch CNN. (d) ResNet. (e) CExNet. (f) IEAE. (g) MAMNet. (h) GT.

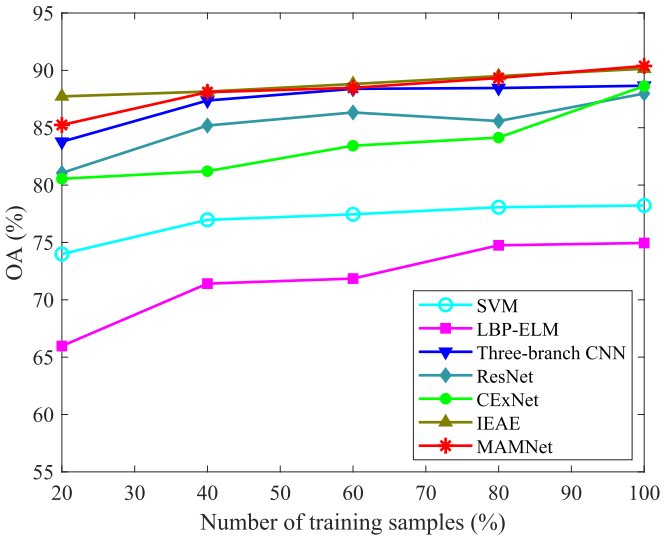


Fig. 8. Relationship between OA and the number of training samples in the Augsburg dataset.

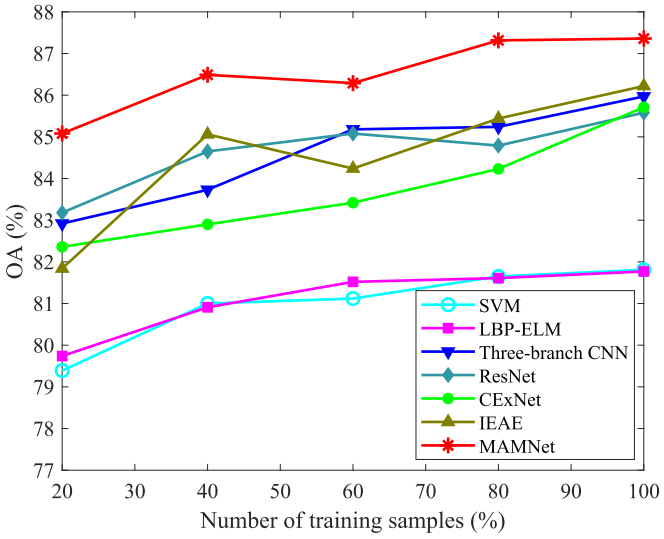


Fig. 9. Relationship between OA and the number of training samples in the Yellow River Estuary dataset.

To sum up, the proposed MAMNet provides a classification map closer to the GT map on both datasets, and it outperforms other competitors greatly in terms of OA and Kappa coefficient. Aiming at some classes with lower classification accuracy, future work will revolve around how to handle classes with small data volumes more effectively.

D. Analysis of Number of Training Samples

Sufficient training samples are the foundation for the model to learn important information until final convergence. However, in the field of remote sensing, the available training samples are usually limited, and labeling these training samples requires a significant amount of time and labor costs. Therefore, it is necessary to propose some feasible methods to enable the model to achieve the highest learning efficiency and classification accuracy even with limited training data. Aiming at this purpose, in some modules of the proposed method, a parameter-sharing strategy is used to reduce the complexity of the model, and data augmentation methods are applied when processing data.

The relationship between OA and the number of training samples of all methods in both datasets is analyzed. Specifically, in the conducted experiments, [20%, 40%, 60%, 80%] of the original training samples are randomly selected to train all the models. The results of the experiments are summarized in Figs. 8 and 9. We can see that machine-learning methods

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are relatively insensitive to the number of training samples. The OA obtained from the three-branch CNN is rather stable when the number of training samples is 60%–80%, but it drops sharply when the number of training samples is 20%–40% on both datasets. The performance of ResNet and the three-branch CNN shows a similar trend with changes in sample size. CExNet is relatively sensitive to the number of training samples, as the number of samples decreases, OA also rapidly decreases. In the Augsburg dataset, the IEAE method is the least sensitive to the number of training samples, but in the Yellow River Estuary dataset, OA drops the greatest when the number of training samples decreases to 20%. In contrast, the proposed MAMNet is less sensitive to the number of training samples, and always achieves relatively high OA among all the competitors.

IV. Conclusion

In this article, a novel MAMNet is designed for pixel-wise multimodal remote sensing image classification tasks. Feature absorption operation, cross-modality attention, and an adaptive tradeoff between diversity and consistency are designed in the proposed network in order to make the proposed model better cope with multimodal remote sensing image information. Specifically, the proposed method benefits from the following three aspects.

1) CIMs are designed to make the extracted features interact with each other to preserve modal-specific information while eliminating redundant information.
2) The MAL is developed to adjust the contribution of each modality to the final feature, thereby enabling the network to better adapt to different scenarios.
3) The AMM loss is calculated to balance the diversity and consistency of multisource features, which can make the network not only learn modal complementary information more efficiently but also enhance universality for various input combinations of modalities. Extensive experiments on the Augsburg dataset and the Yellow River Estuary dataset are conducted, which reflects the effectiveness of the MAMNet even with fewer training samples.

Although the proposed MAMNet has excellent classification performance and high robustness, it may still face new challenges with the further development of remote sensing images. In order to further optimize the proposed method, future work will focus on the following aspects: 1) proposing new strategies to make the model more effective in handling classes with small data volumes and 2) further validating the model by deploying it in larger experimental scenarios.

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